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Assimilation, Accommodation, and the Dynamics of Personality Development

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Source: *Child Development*, Apr., 1982, Vol. 53, No. 2 (Apr., 1982), pp. 281-295

Published by: Wiley on behalf of the Society for Research in Child Development

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Assimilation, Accommodation, and the Dynamics of Personality Development

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BLOCK, JACK. *Assimilation, Accommodation, and the Dynamics of Personality Development.* CHILD DEVELOPMENT, 1982, **53,** 281–295. The present essay specifies some problems in the Piagetian characterizations of assimilation and accommodation and offers an alternative formulation intended to resolve some conceptual anomalies. The dynamics underlying the orthogenetic principle of Werner are derived from this revised and restricted conceptualization of assimilation and accommodation as separable adaptive strategies. Piaget's views on the development of thought are then related to the domain of personality. Equilibration, according to Piaget the "motive" for cognitive growth, is argued to be a manifestation of the individual's orientation to limit anxiety by maintaining or constructing an understanding of the world that is not disconfirmed. This view of equilibrium permits the encompassment of a variety of observations that otherwise seem disparate (e.g., regarding the dynamics of anxiety, the concepts of resiliency or competence, and sex differences).

In psychology, the terms "assimilation" and "accommodation" are strongly and properly identified with the developmental theorizing of Jean Piaget. As conceptualized by him, they are the cardinal processes by which the child constructs and reconstructs perceptual and action schemata for behaving intelligently in the world. Biologically grounded, with extraordinary range, powerfully and elegantly reasoned, Piaget's genetic epistemology has transformed our views of psychological development.

Yet, as with any grand theoretical view, ambiguities, inconsistencies, and gaps remain. The present essay is an attempt to specify some problems in the Piagetian formulation of assimilation and accommodation and to offer a revision to resolve these conceptual anomalies. Consequently, a central feature of developmental progression previously recognized only descriptively (i.e., the orthogenetic principle or law) is explicitly derived. Further, Piaget's notion of "equilibrium" is extended to go beyond simply the cognitive realm into the domain of personality and motivation. This larger view of equilibrium or disturbance permits encompassing a variety of observations that otherwise seem disparate (e.g., regarding the dynamics of anxiety, resiliency or competence, and sex differences). In making this theoretical offering, I respectfully emphasize its dependence on Piaget's magnificent prior achievement. "No behavior, even if it is new to the individual, constitutes an absolute beginning. It is always grafted onto previous schemes and therefore amounts to assimilating new elements to already constructed structures. . . ." (Piaget 1970, p. 707). So it is with scientific ideas as well.

Interpreting Piaget: Problems and Apologies

Piaget wrote prolifically, for an impressively long period, on many complex and interrelated issues. As his views developed, he became his own foremost revisionist, saying things differently over the years—a reasonable kind of inconsistency. The reader should well understand that the task of denoting but also connoting the meaning of even central Piagetian conceptions such as assimilation and accommodation is indeed difficult. Further, the very richness of Piaget's writings admits of differing, perhaps projective, interpretations of the inten-

This study was supported by Research Grant MH-16080 from the National Institute of Mental Health, U.S. Public Health Service. Requests for reprints should be sent to Jack Block, Department of Psychology, University of California, Berkeley, California 94720. I am grateful to Jeanne H. Block, David M. Harrington, Daniel Ozer, Alan Sroufe, Harriet Salatas Waters, and Henry Alker for their friendly criticism of earlier drafts of this article.

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tions underlying Piaget's words. Disciples of Piaget should be prepared in what follows for revisions, of substance or in emphasis, of certain of his concepts.

Recapitulating Piaget's Characterizations of Assimilation and Accommodation

For Piaget, "assimilation is the integration of external elements into evolving or completed structures" (1970, p. 706). Accommodation is "any modification of an assimilatory scheme or structure by the elements it assimilates" (1970, p. 708). Whereas "assimilation is necessary in that it assures the continuity of structures and the integration of new elements to these structures" (1970, p. 707), accommodation is necessary to permit structural change, the transformation of structures as a function of the new elements encountered.

For Piaget, assimilation and accommodation are two sides of the same coin of adaptation. "Cognitive adaptation . . . consists of an equilibrium between assimilation and accommodation. . . . There is no assimilation without accommodation. . . . Accommodation does not exist without simultaneous assimilation either. . . . Only the more or less stable equilibrium which may exist between them . . . characterizes a complete act of intelligence . . ." (1970, p. 708). In such assimilation-accommodation equilibrium, "assimilation is still subordinate to the properties of the objects, or, in other words, subordinate to the situation with the accommodations it entails; and accommodation itself is subordinate to the already existing structures to which the situation must be assimilated" (1970, p. 709).

"If accommodation and assimilation are present in all activity, their ratio may vary. . . . When assimilation *outweighs* accommodation (i.e., when the characteristics of the object are not taken into account . . .), thought evolves in an egocentric or even autistic direction" (1970, p. 708). For Piaget, much of what is called play exemplifies "a kind of free assimilation, without accommodation to spatial conditions or to the significance of the objects. . . . Reality is subordinated to assimilation which is distorting, since there is no accommodation" (Piaget 1951, p. 86); that is, the prototypical example of pure assimilation is play. "Conversely, when accommodation prevails over assimilation to the point where it faithfully reproduces the forms and movements of the objects or persons which are its models at that time, representation evolves . . . in the direction of *imitation*" (Piaget 1970, p. 709; emphasis added); that is, the prototypical example of pure accommodation is imitation.

"In their initial directions, assimilation and accommodation are obviously opposed to one another, since assimilation is conservative and tends to subordinate the environment to the organism as it is, whereas accommodation is the source of changes and bends the organism to the successive constraints of the environment" (Piaget 1954, p. 352). Without assimilation, there would be no sense of continuity, no apperceptive mass—the very fundament of meaning. Without accommodation, there would be no sense of change, no way of escaping a perseverative solipsism. As Piaget remarks, "if assimilation alone were involved in development, . . . [the child] would not develop further" (1970, p. 707).

Piaget presumes, not uniquely, that the organism's neurology has been wired by evolution so as to orient the organism to bring to equilibrium what is not in equilibrium, that is, to equilibrate. "All behavior tends toward assuring an equilibrium between internal and external factors or, speaking more generally, between assimilation and accommodation" (Piaget 1967, p. 103). Further, Piaget suggests that "the construction of structures is mainly the work of equilibration, defined not by balance between opposite forces but by self-regulation; that is, equilibration is a set of active reactions of the subject to external disturbances, which can be effective . . . to varying degrees. . . . The transition from one stage to another is . . . an equilibration in the most classical sense of the word. But since these displacements of the system are activities of the subject, and since each of these activities consists of correcting the one immediately preceding it, equilibration becomes a sequence of self-regulations" (1970, p. 725). Such equilibria are the result, generally a temporary and subsequently succeeded result, of an interweaving of assimilative and accommodative processes according to certain rules deriving from Piaget's "theory of equilibration." This theory is based on two postulates (by postulates, Piaget means "general propositions derived from the study of facts" [Piaget 1977, p. 7]).

First postulate: Any scheme of assimilation tends to feed itself, that is, to incorporate outside elements compatible with its nature into itself. *This postulate assigns a driving force to the process* and therefore must assume activity on the part of the subject, *but by itself does not imply the construction of novelties* [emphases added]; a rather large scheme . . . could assimilate the entire universe

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without being modified or enriching itself in compensation.

Second postulate: The entire scheme of assimilation must alter as it accommodates to the elements it assimilates; that is, it modifies itself in relation to the particularities of events but does not lose its continuity . . . nor its earlier powers of assimilation. This second postulate . . . states the necessity for an equilibrium between the assimilation and the accommodation in order for the accomodation to succeed and remain compatible. . . . Accommodations may be to exterior objects or to other schemes [during reciprocal assimilations][1] and the resultant changes can be exogenous or endogenous. . . . [Piaget 1977, pp. 7–8]

And Piaget adds: "if the second postulate requires the formation of an equilibrium between assimilation and the accommodation, this implies nothing more than: (1) the necessity for accommodations . . . and (2) the conservation of . . . structures when there are successful accommodations. . . . Cognitive equilibrium [is] characterized . . . by mutual conservations . . . attribute[d] . . . to assimilation and the additional processes of accommodation" (p. 8). Piaget further notes, with insight and becoming modesty, that assimilation, accommodation, and equilibrium are notions that "remain on the level of (functional) descriptions." The phrase, "functional description," should not be viewed as a pejorative; rather, the functional descriptors of assimilation, accommodation, and equilibrium represent recognitions of abstract and widely generalizable significance. Whatever specification and close study may be required for precise and vivified understanding of psychological development, Piaget calls attention to the importance of recognizing that the functional goal characterizing all developmental sequences is improved equilibration. And whatever the detailed play of compensations and mutual conservations used by the individual to advance toward the equilibration goal, Piaget calls attention to the importance of recognizing that the overarching functional strategies of assimilation and accommodation are involved.

Some Conceptual Problems with the Notions of Assimilation-Accommodation and Some Suggestions

Evocative and implicative though these Piagetian formulations have been, I suggest they are now limiting of efforts to more fully specify the sequence of both short-term and long-term developmental change. What are some of my concerns?

1. Piaget seems to posit the co-occurrence of assimilation and accommodation. Such simultaneity, if indeed conceptually intended by Piaget, would seem to foreclose on the possibility of close analysis of the interplay or sequencing of assimilation and accommodation. Logically, the idea of co-occurrence implies an interval or category imposed upon a time continuum or a directional sequence. The breadth or narrowness of the interval or of the category can entail simultaneity or an absence of simultaneity of the events or processes under consideration. Thus, if an arbitrary time unit of 24 hours is employed, then day and night must be said to co-occur; if the time unit employed is 1 hour, then day and night are distinguishable, and sequential relations between light and darkness may be discerned. Of course, it is always feasible to go from a more finely resolving partitioning of experience to a less finely resolving partitioning, by simply relaxing certain criteria. But it is not possible to go the other way. Timing by hourly units can permit timing by days, but timing by days denies the possibility of discriminating by the finer hour. With reference to assimilation and accommodation, I suggest that it will be helpful in going beyond their current usage as only (and amorphous) functional descriptors of equilibrative acts to make explicit conceptual use of their separability as processes.

Certainly there is an interplay of assimilation and accommodation, and this interplay is often rapid—so rapid perhaps as to suggest simultaneity if the analytical unit is not fine or only a quick and approximate characterization is intended. But the interplay of two processes does not mean simultaneity. A scheme or mental representation includes structural parameters which can take on various values according to the circumstances the individual is in. The individual tries to assimilate objects or events to the established parameters of a scheme by arriving, easily or with difficulty, at a set of tenable parameter values that can generate an adaptive perception or action. Only when adaptation is not feasible by changing parameter values does the individual shift over to changing the parameters themselves, that is,

[1] A recent definition of reciprocal assimilation: "when two schemes or two subsystems (for example, looking and seizing) are applied to the same objects or coordinate without having further need of the actual content. We can even consider as a reciprocal assimilation the relations between a total system, characterized by its own laws of composition, and the subsystems which it includes with their differentiations, for the subsystems' integration into a whole is an assimilation to the common structure" (Piaget 1977, p. 6).

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to accommodation of the scheme. The adaptively oriented changing of parameter values (assimilation) and the adaptively oriented changing of parameters (accommodation) cannot be coincident; they must occur sequentially.

In my preferred terms, then, the adaptive efforts of the individual involve an interweaving of the separable processes of assimilation and accommodation; a recursive nesting of these separable strategies for equilibration; the alternating application of these separable programs for adaptation that, via a succession of diminishing overcorrections, "hunt" for and "center" on reality or a sufficient equilibration. This way of looking at assimilation and accommodation is a change in emphasis from Piaget's views and has implications to be developed later in this essay.

2. Piaget employs two different models for conceptualizing the processes of assimilation and accommodation, a physiological model and a psychological model. However useful the physiological model may be for conveying certain ideas intuitively, it is a source of confusion or misdirection or even paradox, I suggest, when the psychological model is also considered. Thus, in employing the physiological analogy of digestion as a basis for characterizing assimilation and accommodation, Piaget appears to mean that the organism *actively assimilates* objects and *passively accommodates* to unassimilable objects. But when Piaget employs a more psychological model, he describes ordinary life events as being processed almost automatically (and passively) by assimilative adaptive structures and fitted to existing schemes. In the psychological model, extraordinary events—events that are, in Piaget's terms, "intrusions"—cannot be processed automatically because existing adaptational schemes do not suffice. The individual must actively seek—that is, construct and reconstruct—schemes that can encompass the intrusion, equilibrating it with past and other events by forming a newly sufficient organization of the domain of experience involved. Thus, accommodation, when defined involved. Thus, accommodation, when defined tion or conformation to the "particularities of events" (Piaget 1977, p. 7) but an active, constructive, "go[ing] beyond the present state to seek new equilibriums" (p. 12).

The contrary implications of the physiological as compared to the psychological conceptualizations of assimilation and accommodation seem to me to have led to an important and widely held misunderstanding of the nature of play and imitation. More than a little, this misunderstanding is attributable to a failure to read Piaget closely.

Reasoning from the physiological model, Piaget offered play as the prototypical illustration of assimilation and imitation as the prototypical instance of accommodation. In characterizing play as an egocentric "free assimilation," Piaget was referring to a special kind of play at a special time of life. His primary reference was to the play of "fictional games, in which objects at [the child's] command are used only to represent what is imagined . . . [and which] is most frequent at the beginning of representation [1½ and 3 years of age]" (Piaget 1970, pp. 708–709). Such emphasis on assimilation is distorting and indicates "the subject remains centered on his own actions and his own viewpoint" (Piaget 1970, p. 710).

However, although fully egocentric assimilation occurs (as in dreams), observation suggests that play, even in the fictional games of young children, is firmly rooted in this world. Within the psychologically safe context provided by an egocentricity that escapes or delays confrontation with the external world, the playfully fantasizing child appears to function in active, experimenting, metaphorical, preliminary mastery, belated mastery ways that presage structure formation and structure modification, that is, accommodation. Piaget recognized this aspect of play in his observation that fictional games "evolve toward constructive games in which accommodation becomes more and more precise" (1970, p. 709). Thus, it would appear that the active, experimenting, constructive function of play as the child acts on and receives feedback from the world being experienced—by no means an autistic, solipsistic one—is neglected in the usual equation of play (unqualified) with extreme assimilation, if assimilation is *psychologically* defined as the utilization of existing structures for the organization of experience.

In characterizing imitation as prototypically accommodative, Piaget again was referring to a special kind of imitation at a special time of life—imitation through action of the models that happen to be present when the capacity for representation is achieved.

However, although the imitation of a young child "faithfully reproduces form and movement" and hence may be "pure accommodation" in the physiological sense, it is also the case that such emphasis on accommodation is distorting and indicates, to paraphrase Piaget,

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that *the imitating child remains centered, albeit on another's actions and another's viewpoints.* By its usual definition, the concept of imitation implies that an external structure for guiding perception or cognition or behavior has been incorporated, generally abruptly and without modification, by the individual as his internal structure for shaping experience and behavior in an equilibrating way. That is, imitation means rote knowledge and not evolved, cognitively registered schemata or schemes that can be operated upon. The imitating child has incorporated prescriptions and proscriptions from outside regarding how to perceive and act in the world. But the imitating child will not know the when or the why of the behaviors being emitted and will be nonplussed by circumstances with which the internalized structures for imitation do not mesh. Such imitation may serve an immediate or local equilibrative purpose for the child, but the imitating child's knowledge is figurative rather than operative. If viewed in terms of the psychological model, imitation seems passive, "mere copying," and unlikely to lead to "inventive constructions" (1970, p. 714) by the individual that will permit subsequent accommodative competence and resourcefulness.

Much depends on just which model of assimilation and accommodation is employed. In my view, the alimentary analogy of Piaget should be set aside in favor of the more complex psychological conceptualization that also is to be found in his writings. Specifically, I suggest that the psychological concept of assimilation be defined as and restricted in meaning to the utilization of existing adaptive structures or schemes to integrate or make sense of new experiential elements, and that the psychological concept of accommodation be defined as the formation of new adaptive structures or schemes to integrate or make sense of previously discrepant experiential elements. By so doing, the inventive constructions and the metaphorical cognitions that so frequently typify play become, conceptually, instances of accommodation; the rote compliance so frequently characterizing imitation can be seen to be assimilative (albeit by the invocation of adaptive structures of external origin).

It should be noted that this proposed revision of the classification of play and imitation as prototypical instances of assimilation and accommodation does not commit the converse error of positing play as "pure" accommodation and imitation as "pure" assimilation. Whether play or imitation involves accommodation (or assimilation) may have to be evaluated on a case-by-case basis employing, as the criterion, whether (or not) internal structural change of the child's concepts is involved. I do suggest, however, that a psychological rather than a physiological perspective on assimilation and accommodation better appreciates the roles and relative significance of play and imitation in developmental change and competence enhancement.

Finally, the restriction of the concepts of assimilation and accommodation to a purely psychological definition also means that what Piaget has called "reciprocal assimilation" must be seen as an instance of accommodation.

3. Cognitive structures are disequilibrated by intrusions, that is, real or imagined discrepancies between the environment and the individual's cognitive schemes for encompassing or "processing" that environment or discrepancies between two cognitive schemes. From Piaget, there is the suggestion that, in such circumstances, accommodation processes inevitably come into play as the individual seeks to equilibrate or reequilibrate. While such a position is not incorrect if the notion of accommodation is employed only as a functional descriptor, if more specific conceptions of assimilation and accommodation are employed, different implications follow. Although intrusions do indeed present an impelling problem for the individual, such intrusions need not dictate the nature of the compensation that will (at least partially or temporarily) equilibrate the individual. In particular, such intrusions need not elicit accommodation as the inevitable or appropriate compensating activity.

In response to such intrusions, the compensating individual, if narrowly and rigidly egocentered, may reinvoke the strategy of assimilation in an insistent, intensified, even procrustean way to reequilibrate. But this reequilibration will be of an attenuated, restricted form. Other individuals, also egocentric but artistically autistic, may achieve a perpetuation of their assimilative structures via creative redefinitions and simplifications of the disequilibrating intrusions.

The alternative response to intrusions requires the individual to make what ultimately is a developmentally more progressive move from egocentricity toward a wider and more flexible perspective on the world, that is, the individual may employ the strategy of accommodation as a means of forming new or different compensations that move toward equilibra-

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tion. Such accommodations may involve new, meaning-making structures that build upon previous assimilative foundations, or they may involve structural transformations of a more fundamental nature. Creativity of a high order may be manifest in these accommodations. But also, accommodations may be so quick and endlessly sliding as to suggest an absence of structures, a characterlessness, as the individual thermoplastically conforms to the hot experience of a discomfiting situation.

However, whether conceived in positive or negative terms, assimilation and accommodation can be viewed as distinguishable and alternative processes serving the individual's preprogrammed goal of achieving or at least advancing toward a coherence-making equilibrium.

4. In Piaget's theorizing, the meaning of an equilibrium between assimilation and accommodation is unclear. It is rhythmic but not revelatory to define such equilibrium as simultaneously involving assimilation that is subordinate to the properties of objects and accommodation that is subordinate to already existing structures. Piaget developed his views most fully with respect to the equilibration of cognitive structures; he was vague or terse about the equilibrium of assimilation and accommodation. For Piaget, equilibration involves compensations or self-regulating activities by the subject to respond, correctively, to external intrusions. This conception of a mobile equilibrium ("homeorhesis"), a dynamically responsive set of equilibratory arrangements evolved by the individual, requires description of the "play of compensations" by which equilibrium is achieved or at least moved toward. With respect to the development of cognitive structures, Piaget (1977) has provided a characterization, often detailed or entailing, of the equilibrative process. But the idea of an assimilation-accommodation equilibrium was only asserted by Piaget and never given specificity. Certainly, the result of the interplay of assimilation and accommodation is improved equilibration. But an improved equilibration does not mean that, in its achievement, the processes of assimilation and accommodation are themselves in equilibrium. I suggest, therefore, that it may be useful to set aside Piaget's emphasis on an unspecified and theoretically unused equilibrium between assimilation and accommodation.

In its place, I suggest, in a more explicit and stronger form than can be found in Piaget's writings, that, through the course of evolution, individuals have been programmed to follow the adaptive imperative: "Assimilate if you can; accommodate if you must!" Assimilatory efforts are the first line of adaptation. When assimilation works or can be made to work, it is adaptively economic, preserving existing equilibrations of cognitive structures and maintaining the individual in a tolerable or sufficient state. It is only when assimilation fails, for whatever reason, to reconcile intrusions within the structural status quo that the destructured individual attempts the second line of adaptation, accommodation. Whereupon the formation of new structures, if effective, can be used to assimilate what before was unassimilable, thus again reequilibrating the individual. As Piaget emphasized many times, the repeated interplay of these meaning-making strategies provides a heuristic for adaptation of profoundly wide and profoundly deep applicability.

To summarize the conceptual suggestions made thus far: assimilation and accommodation, instead of being defined as inextricably fused, have been definitionally separated. A mixing of theoretical metaphors in Piaget's writings has been argued by me to be confusing and misleading, especially in the way play and imitation often are construed. Restricted (but also I believe more specifically implicative) definitions of assimilation and accommodation as alternative strategies of adaptation have been offered. Assimilation is to be viewed as the invocation by the individual of existing adaptive structures, schemes, or scripts to process experience; accommodation is to be viewed as the shiftover by the individual to the formation of new (and the re-formation of old) adaptive structures, schemes, and scripts to process experience. Schematizing (assimilation) and schemata modifying (accommodation) are very different adaptive modes. The motivational functions of both assimilation and accommodation are to advance or to restore psychological equilibria, to equilibrate or to reequilibrate. The defining feature of equilibration is the integration of otherwise discrepant experience, understandings, and actions. Assimilation and accommodation, separately and in helical, perhaps quite rapid but also perhaps quite slow alternation, serve to equilibrate the individual. Piaget's asserted but unused idea of an equilibrium between assimilation and accommodation has been set aside. Instead, it is posited (as Piaget sometimes suggested) that the assimilative mode has a built-in priority over the accommodative mode, so that the individual confronted with discrepancy first will seek to equilibrate via assimilation, moving on to the accommoda-

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tive mode only when, given the individual's existing structural parameters and limits, assimilative efforts have not achieved sufficient integration of the disparate and regnant elements requiring understanding.

Deriving the Orthogenetic Principle of Development

To illustrate the explanatory possibilities conferred by the preceding restatement of the nature and connection of assimilation and accommodation, consider the orthogenetic principle or "law" of Werner (1948, 1957), so frequently cited as characterizing psychological development. The orthogenetic principle, as usually understood, asserts that, with normal maturation, individuals become both more differentiated and more integrated in their behavior. "Wherever development occurs it proceeds from a state of relative globality and lack of differentiation to a state of increasing differentiation, articulation, and hierarchic integration" (Werner 1957, p. 126). Or, "under the widest range of conditions, organisms undergo transformations from the status of relatively undifferentiated entities to relatively differentiated and integrated adult forms" (Werner & Kaplan 1963, p. 5).

Observationally and intuitively, this principle seems generally true. But, although a useful observation, the orthogenetic trend characterizing individual development has remained only a descriptive recognition, what Werner calls "a heuristic definition." The sequential dynamics underlying such orthogenesis have not yet been proposed.

Within the conceptual framework just argued, the orthogenetic law would be derived as indicated in the following verbal flow chart:

a) During the early course of development, we may designate the individual at a particular time (say, stage *n*) as assimilating experience in a tolerable way; that is, incoming perceptions and outgoing actions are integrated and guided by such schemata as then exist.

b) Because the individual lives in a world providing at least some new forms of experience, the normative, biologically expectable individual inevitably and continuously will become more differentiated with the passage of time. By differentiation, I mean (not uniquely) the increasing articulation of experience or action, the continued aggregation of simple learnings of (unconnected) perceptual and action elements, the further psychological noticing or separation of what previously had gone unnoted or unseparated, the accretion of what Piaget labeled "exogenous acquisitions" (1970, p. 713), learnings that have not been structured or organized. Although the rote learnings that result from the process of increasing stimulus and response differentiation may, and often do, have direct and immediate adaptive implications via the contingencies of reinforcement, the existence of differentiation alone—without structure or organization—permits only or rather requires "a systematic [I would add, passive] dependency of the subject on the object" (p. 713) and on the noticed contingencies in the world beyond the subject. Nevertheless, the reflexively reactive, stimulus-response perceptions and actions consequent on differentiation alone may often, for some individuals and/or for some times and contexts, be directly adaptively sufficient and thus terminate further efforts to improve on adaptation. Under such circumstances, Werner's orthogenetic progression will not be manifested—there is differentiation, but without integration.

Generally, however, with enough passage of time in a complex, changing, unpredictable world, and given the neurological capacities of the human being, the differentiation of an individual will increase and can be expected, with great probability, to impose appreciable adaptational problems. This further differentiation of the individual may proceed in two kinds of developmental contexts, where there are no (or few, or very simple) integrating or schematizing structures available to the individual (see step *c* below) and where the individual has some (appreciable and developed) integrating and schematizing structures (see step *d* below). The line would be difficult to draw between the idealized circumstances where there are none and those where there are some integrative perceptual and action structures. Indeed, ethology suggests evolutionary reasons for presuming that the first case may not exist and that organisms begin life with *some* integrational structures, albeit perforce primitive or only broadly orienting ones. Not fortuitously, identical adaptational consequences finally follow from increasing differentiation, whatever prior integrational structures are or are not available to the individual.

c) Presuming no or few integrational structures are available, extended but unorganized differentiations would accumulate and come to overburden the intrinsic information-processing (encoding, storage, and retrieval)

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capacities of the individual. That is, beyond certain memorial limits, increasing differentiation can be expected to pose new and compelling adaptational problems for the individual because the memorial weight of continuing differentiation is heavy. The registering individual can become overwhelmed and ineffective when confronting the massive and inundating tasks of encoding, storing, and retrieving the information contained in the many "exogenous acquisitions" or stimulus-response connections that complicated experience brings.

However, by the fortune of their neurology, human beings have available to them, besides the narrowly algorithmic tactic of differentiation followed by the association of specific responses to specific stimuli, the broad and heuristic cognitive strategy of schematizing or integrating what they have differentiated. The schematizing or integrating of what has been differentiated, the move "from association to structure" (Mandler 1962), provides cognitive economy, of course. And, if categorical incisiveness is achieved (Rosch 1978), there need be little or at least acceptable discriminative cost. Thus, a compelling adaptive solution for the problems posed by increasing differentiation is the formation of integrating or schematizing structures where they did not exist before, that is, accommodation.

The construction or achievement of these new integrative structures as a function of the information-processing burdens consequent on increasing differentiation provides a beginning instance of the orthogenetic principle—of a progression from stage n to stage $n + 1$ where stage $n + 1$ is characterized both by greater differentiation and by greater integration. The individual now is back at step *a*, above, but at a higher developmental level, and the developmental cycle repeats, through step *b* and on to step *d* where, now, increasing differentiation occurs in the context of prior schemata or integrating structures.

d) Where prior schemata or integrational structures exist, with continued experience of a complex, changing, unpredictable world, increasing differentiation can be expected to present the individual with experiential elements that mountingly confront or embarrass or disequilibrate existing assimilative schemes.

In response to the faltering effectiveness of assimilative structures, dedifferentiation may occur, thus permitting renewed and perhaps now successful efforts to assimilate. By dedifferentiation I mean the reversal of previous differentiation, the failure to notice or to keep separate what previously had become noticed or separated. The tactic of reinvoking an already failed assimilative schema, when aided by dedifferentiation, prevents the psychological existence or registration of the disparate elements threatening the assimilative scheme and places the individual back at step *b*, whereupon the individual soon progresses again to step *d*. If the capacity for dedifferentiation does not exist or its limit on excluding experience has been reached, the individual no longer can in step *d* call upon further perseveration of the wanting assimilative mode as an adaptive resource.

e) At this critical juncture of sufficient mismeshing of the implications of the newly differentiated recognitions vis-à-vis the established conserving or integrative assimilative schemes, passive assimilation along previous lines is abandoned. The individual now shifts over to active accommodation, to establish a new scheme that can reconcile the newly differentiated experience and actions available to the individual with the experience and actions previously organized or equilibrated by the proved insufficient and therefore abandoned schemes. The newly achieved accommodation, if effective, will thereby manifest a higher form of integration than the previous assimilative scheme because it encompasses the differentiations that prompted the shift from the inadequate assimilative mode to the formation of an adequate assimilative mode. This new, reconciling, mastering scheme now functions to assimilate ongoing experience in a tolerable way. That is, incoming perceptions and outgoing actions are integrated and guided by the now-existing integrating scheme. The individual is back at step *a*, above, but at a higher level (stage $n + 1$) both of differentiation and of integration. Q.E.D. The developmental cycle now repeats.

Several implications or aspects of this account of the sequence by which orthogenesis proceeds warrant attention. When plotted against time, level of differentiation would be shown by a continuously rising curve, whereas level of integration would be represented by a staircase or step-function or discontinuous curve. That is, during the orthogenetic course, level of integration tends for a while to be relatively stable, thus falling behind the rising level of differentiation, and then catches up in an abrupt and large rise. Further, one can expect a decrease or dip in level of integration during the interval between abandonment, after failure, of assimilative modes and the implementation of an effective accommodation that now

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can assimilate. This period, usually but not necessarily brief, would be manifested in a dip in the staircase curve, just before the riser begins. The existence of this dip signaling destructure or disintegration prior to the rise of new integrating structures has been remarked upon earlier, and in rather similar terms, by Werner. "One has to regress in order to progress . . . [one form of psychological regression involves the] dissolution of existing, schematized or automatized behavior patterns" (1957, p. 139). (I note that Werner uses the term dedifferentiation to refer to the dissolution or what I call the destructuring or disintegration of previously organized perceptual or action schemes. As will be recalled from step *d*, I use the term dedifferentiation to refer to the unavailability in immediately effective memory of previously acquired simple and unorganized learnings. These are very different terminological usages and should not be confused. Whereas I view dedifferentiation as serving, albeit often ineffectively, an equilibrative function, I view destructuring as a sign of disequilibration.)

The orthogenetic sequence outlined articulates a cyclic process and a dynamic by which development proceeds. The sequence applies to the process of short-term change within a developmental stage, as when the individual enriches or embroiders upon a basic assimilative structure by a number of small, rapid, and local accommodations (such as go on when the infant learns to grasp and to grope). And it characterizes as well the process of long-term structural change, of transition from one developmental stage to another, when basic assimilative structures are transformed by large, slow, far-reaching accommodations.

I also suggest that, as many individuals lead their lives after establishing a modus vivendi, the orthogenetic sequence may lose its pertinence. Once having found stable, circumscribed, and "satisficing" (Simon 1955) niches for living, many individuals may not become more differentiated or require further revision of their assimilative structures. The passage of time need not mean development.

As is widely recognized, the problem of transition from one great developmental stage to the next has not been explained in a satisfying way by Piaget and other stage theorists; the preceding flow chart may be a more attractive and specifying characterization of the sequential dynamics involved than the somewhat mysterious explanations sometimes offered in the static terms of a dialectical confrontation and an announced emergent resolution.

The Relation between Piaget's Views on the Development of Thought and the Domain of Personality

The broad concepts of assimilation, accommodation, and equilibrium as restated here have implications going well beyond simply the description of cognition; they can also be applied to conceptualize the dynamics of personality development and motivation, topics usually avoided by Piaget. As with cognitive development, personality development represents a constructive achievement by the individual. Although biologically given constraints and shapings exist, the personality structure of an individual is not therefore preordained. Nor is the personality structure of an individual a simple, template-formed organization of perceptual, cognitive, and action schemata and schemes provided by the larger social context. Rather; the personality system of an individual is: (*a*) a more or less idiosyncratic construction (*b*) of dynamically linked, equilibrating ways of motivation processing (*c*) predicated on a set of orienting premises (*d*) regarding a complex, largely interpersonal world (*e*) where communication and feedback regarding the effects of behavior can often be imprecise or even nonexistent.

For an understanding of how these personality ways of achieving or maintaining or advancing a motivational equilibration come about, many features of Piaget's developmental principles for understanding cognitive development can be largely adopted and adapted. For Piaget, however, intelligence grows within the orderly, relatively manageable, and responsive-to-behavior physical realm. To paraphrase Klaus Riegel (1978, p. 104), Piaget's analysis has been restricted to the interaction between active organisms and passive objects. In contrast, the personality system evolves in the relatively unmanageable and unresponsive-to-behavior, fuzzily defined, projectively interpretable, affectively involving interpersonal realm. In this more usual and more motivating world, largely neglected by Piaget, both the individual and the objects are active, reciprocating participants.

For this reason, Piaget's principles, although helpful, cannot provide a sufficient basis for conceptualizing personality and personality development. The difference between a passive and passionless world about which one can generate unequivocal understandings and an interactive, passion-involving world where understandings never rise above uncertainty or am-

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biguity is a profound one. The principles of cognitive growth useful for the first world, where the improvement of cognitive equilibration seems to provide its own impetus and external criteria can be referenced, can be only partially useful or relevant in the second world, where motivations are compelling and external criteria very often are lacking or unclear. New principles of psychological adaptation must be constructed by the individual that can equilibrate, at least sufficiently, in this second, wider, and more usual world. These principles will take the form of perceptual, cognitive, and action structures or schemes (premises, orientations, attitudes on the world, deuterolearnings, scripts, etc.) that are motivationally equilibrative with little or no, or ever construable external or independent corroboration.

Cognitive psychologists sometimes forget that *the* biosocial problem of the individual is adaptation. Although Piaget fused and even equated intelligence with adaptation in such phrases as "adaptive intelligence" and "intelligent adaptation" (Piaget 1954), there are ways of adaptation that do not use intelligence in the Piagetian sense of the capacity for logical reasoning derived from and applicable to experience of the physical world. Although the capacity for intelligence in these terms has been biologically evolved as a most powerful means of adaptation, it is adaptation that is the key, and intelligence is but one way. Other adaptive structures exist (e.g., reflexes of various kinds, genetically provided fixed or modal action patterns promoting the adaptive strategies of perseveration and of exploration) or are achieved by the individual (e.g., personality mechanisms such as paranoia or hysteria, providing biologically driven adaptive resolutions). To the extent that intrapsychic construals or redefinitions or action patterns are formed within the individual as adaptive (if formally irrational) solutions or strategies, and these go undiscorroborated and are interwoven with other means of adaptation (e.g., intelligence), it is necessary to recognize the very large domain of personality. Within that domain, intelligence and much of what is viewed as cognition can be placed as providing a means—an extraordinarily effective means often, but not the only means—of achieving or enhancing biosocial adaptation.

Some Personality and Motivational Implications of These Revisionist Views

Within this perspective on the proper definition or delineation of the domain of personality psychology, I suggest now some connections between the Piagetian notions earlier discussed and certain aspects of personality functioning.

1. Piaget postulates cognitive disequilibration as intrinsically motivating, "a psychic need *sui generis* [in its own right] . . . the principal motive power of intellectual activity thus becoming the need to incorporate things into the subject's schemata" (Piaget 1952, pp. 45–46).

It would be more attractive to derive this intrinsic motivation as a theorem than to simply presume it. Toward this end, consider that cognitive disequilibration is also an indication to the individual that the world, or a portion of it, does not make sense or have meaning and is therefore not predictable or manageable. When viewed from the overarching and compelling framework now set by evolutionary biology, the adverse adaptive implications of disequilibration are clear. It therefore seems reasonable to view a registered cognitive disequilibration as a signal to the individual, an alerting to a survival threat however indirect or far removed. And such signals or alertings, by the screenings of evolution, can be expected to have assumed both dysphoric qualities and motivating properties, that is, to be anxiety inducing.

As a more immediate rationale for considering cognitive disequilibration to be an instance or manifestation of anxiety arousal, consider the long recognized connection between anxiety and the interruption of behavior as discussed by Freud, Lewin and others, and most recently in the seminal writings of Mandler (1964, 1972, 1975, 1980; Mandler & Watson 1966).

Observes Mandler (1980, p. 234):

> The phenomena that have been phenomenally and theoretically labeled as anxiety . . . [arise] out of the combination of the interruption of ongoing behavior and thought on the one hand and the unavailability of appropriate, situation-relevant task- or problem-solving actions on the other. The interruption is . . . assumed to lead to the arousal of ANS activity and represents the visceral component of the anxiety reaction. The unavailability of situation-relevant actions or thought [Mandler labels as the condition of] helplessness; the individual is faced with a situation [either in the external or the internal world] that he or she cannot adequately address—The anxious individual does not know what to do in the problem-solving sense. Interruption accompanied by a lack of appropriate problem-solving actions or thoughts is represented in private experience as a subjective emotion that is frequently categorized as anxious [or nervous, or tense] in the common language.

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Mandler brings together diverse, and in my view compelling, evidence and reasoning supporting the conceptualization of anxiety or helplessness or distress or dysphoric arousal as resulting from the individual's disorganization due to the interruptions of previously flowing sequences, frustrations of actions, discrepancies between expectation and actuality, and unanticipated conflict which causes ANS arousal that is not under the control of the individual. And, of course, to say the interruption of an integrated perceptual-cognitive-response sequence creates a state of dysphoric arousal is simply another, but more encompassing, way of characterizing what in Piaget's terms would be called a failure of assimilative schemata with resulting cognitive disequilibration. In this larger perspective, then, disequilibration would result in feelings ranging from unease through distress to anxiety. And such dysphoric arousal would be motivating—conditions or behaviors or strategies associated with a reduction in such arousal would be reinforced. Equilibration, *however achieved*—whether by an act of intelligence or by a fortuitously reinforced behavior or by an act of reality-distorting defense—may therefore be coordinated with a reduction in dysphoric arousal and, at a phenomenological level, a lessening in anxiety, the experience of helplessness, distress, and unease.

By conceptualizing the tendency of individuals to cognize not as a motivation sui generis but as deriving from an ingrained and adaptive strategy to maintain the state of anxiety within bounds, a theoretical economy is achieved, and questions regarding "hot" and "cold" cognitive development (Kessen 1971) and hot and cold cognition (Nisbett & Ross 1980) become less polarizing. The field of cognitive psychology and information processing becomes no less important when placed within a broader, biologically influenced framework. Indeed, some of the problems cognitive psychology seems to be having in wrestling with the difficulties of context and meaning and the schematic bases of social information processing (Taylor & Crocker 1979) may be simplified by the recognition that the individual is pulled or pushed by a personal criterion only imperfectly related to the external world.

2. Assimilation and accommodation are viewed by Piaget as strategies to maintain, restore, or advance equilibration. If equilibration is coordinated with a reduction in anxiety, then assimilation and accommodation may also be viewed as strategies to prevent, undo, or decrease the anxiety looming when equilibrative modes falter. "Making the world make sense" is the criterion for a successful assimilation or accommodation. The world makes sense when the world is predictable or at least not disconfirming of expectation. The key role of predictability derives from its immediately adaptive and ultimately reproductive significance. In its logically weaker but generally sufficient psychological version, the goal of predictability reduces to the attainment of premises or schemes that cannot be or tend not to be disconfirmed.

3. Assimilation, the use of existing structures for the organization of experience and action, is the first line of adaptation (or defense) for the individual in attempting equilibration. When assimilation fails, the individual is destructured or disorganized, that is, is without effective assimilative structures. Thus disequilibrated, the individual is dysphorically aroused. To the extent an assimilative structure is central or fundamental to the individual's adaptive personality or cognitive system, the disequilibration resulting from its insufficiency and failure can be expected to result in intense and agitated dysphoric arousal.

4. In response to such passively experienced anxiety states due to destructure, the individual is drawn to actively formulate a structure sufficient to encompass what was not encompassable before. He may, via effective accommodation, construct or invent new schemes that are equilibrating and can be used subsequently to assimilate the previously unassimilable. Be it noted, however, that the route to a mastering accommodation may entail prolonged enduring of a state of destructure and even further demolishing of unworkable assimilative structures before the construction of a new or reformed accommodative scheme can begin. This extension and completion of destructure will intensify the already present dysphoric arousal; the individual must live with anxiety during the time a creative accommodation is being worked through.

5. If, for the particular individual involved, the anxiety or unease experienced while new accommodative modes are being evolved cannot be endured because that individual's personal limits or tolerance for anxiety is being exceeded, the individual may return, perseveratively, to assimilative efforts. Failed integrational structures may be reinvoked and perhaps made to work by the developmentally regressive mechanism of dedifferentiating (i.e., removing) the scheme-contradicting elements demanding equilbration.

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In effect, instead of solving the equilibrative problem presented by the confronting discrepancies between experience and scheme (or two mutually contradicting schemes), the problem is belatedly and retroactively prevented from arising and disturbing the ongoing individual. The hysterical style of personality (Shapiro 1965) exemplifies this adaptational tactic.

6. Holding degree of differentiation constant and presuming that the assimilative mode has a priority, built in by evolution, over the accommodative mode, we may distinguish individuals with respect to their readiness or reluctance to switch from the assimilative mode of processing experience to the accommodative mode when disequilibration arises. In reacting to new experiences failing to conform to the expectations generated from existing schemes, some persons are able to (and even insist on so doing) encompass the new information in ways that require little or no modification of previously established premises. Other persons encountering new experiential inputs have a greater readiness (and even preference) for leaving schema that do not mesh readily with the new information and to move on to the formulation of new and sufficient meaning-making structures. The ordering of individuals with regard to their degree of continuation with insufficient assimilative structures before attempting accommodation represents, within the present perspective, a primary conceptual variable for the study of individual differences.

Prolonged assimilative efforts can be expected to be manifested as perseveration, an absence of the registration of discrepancies, "too enthusiastic" an application of schemes (Taylor & Crocker 1979), and even an "intolerance of ambiguity" (Frenkel-Brunswick 1949). Shifting to an accommodation too readily can be expected to be manifested in behavioral fluctuations, numerous and ever-changing perceptual-cognitive-action recognitions of possibility, and even what might be called an intolerance of simplicity. The characterological qualities of individuals at the two ends of this continuum are nicely elicited by studies of "leveling-sharpening" and by the Bavelas "completion of squares" task employed by Crutchfield (1951) which evaluates the comparative ease with which individuals break up a problem-solving structure that is narrowly valid but is incompatible with the problem solution defined in a broader sense.

From an adaptational perspective, neither prolonged imposition of insufficient assimilative modes nor flighty shifts to transitory accommodative solutions are effective as general strategies. For particular, happenstance situations, one or the other extreme adaptational styles may prove fortuitously to be an extraordinarily pertinent tactic. But in the heterogeneous mix of circumstances an individual may expect to encounter, the adaptational race can be expected to go to the individual who shifts over to accomodation from assimilation neither too late nor too early.

Earlier, individuals who manifest these apposite abilities have been called ego-resilient (Block 1965; Block & Block 1980), a term intended to imply a general adaptive resourcefulness in contrast with an "ego-brittleness." The popular construct, "competence," in its larger sense (White 1959) rather than in its multifarious, context-specific meanings (Anderson & Messick 1974), also seems to me to be conceptually understandable or derivable as a function of the individual's personal changeover point from assimilation to acccommodation. The present formulation thus may provide a more conceptual and dynamic characterization of just what underlies the labels of ego resiliency and competence. But before the fruitfulness of this conceptualization can be evaluated, explicit and specific tests of these conjectures will be required which employ procedures and measures designed to order individuals on the assimilation-accommodation continuum. Some of this research is already under way (Block & Block 1980).

7. The idea that individuals can be distinguished reliably and interestingly with respect to their tendencies to maintain assimilation as compared with their readiness to shift over to accommodation has been employed by Jeanne H. Block (1979; Note 1) as a possible conceptual basis for understanding many of the findings of psychological differences between the sexes.

As already observed, individuals may overassimilate (procrustean perseveration) or they may overaccommodate (change in complaisant rather than integrity-preserving ways); they may assimilate creatively or they may accommodate creatively. No value judgment per se can be placed on the adaptive strategies of assimilation and accommodation. Within the context set by this recognition, it is of interest, however, to consider the currently differential socialization of the sexes in this society.

An integrative review of the literature on the differential socialization of girls and on boys

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(Block, Note 1) concludes that females grow up in a more structured and directive world than males. Girls experience more parental supervision, more restrictions on exploration, more household chore assignments, more emphases on maintaining proximity, and more frequent (and sometimes unnecessary) help in problem-solving situations. All of the sex-differentiating influences combine to create a more canalized and more predictable environment for girls than for boys, whose encounters with the world outside the home are both more extensive and less controlled. Contributing also to the more predictable world of girls is the nature of their preferred (imposed?) play activities, which, more than the play of boys, involve model availability, instruction, and rule governance (Fagot 1973; Lever 1976; Carpenter, Note 2). At older ages, the games preferred by girls continue to be more rule bound and make fewer demands for improvisation than the games of boys, which often require extemporaneity and the ability to be opportunistic in a momentarily advantageous situation. Since, when sex is held constant, relatively structured environments have been found to elicit less innovation and more compliance in children (Carpenter, Note 2), to be associated with less task engagement (Fagot 1973), and to be related to intolerance of ambiguity (Harrington, Block, & Block 1978), these formal differences in the learning environments provided to girls as compared with boys can be expected to have cumulative, powerful, and general effects on the adaptive strategies invoked at times when expectations fail.

Block (1979; Note 1) has therefore conjectured that girls, more than boys, are socialized in ways encouraging the use of assimilative strategies for processing new information, while boys, more than girls, are socialized in ways that encourage the use of accommodative strategies when confronted with informational inputs discrepant with prior understandings. As a corollary, girls, more than boys, can be expected to be slower to leave efforts to assimilate for efforts to accommodate. Socialization practices creating relatively structured environments, more common in the case of females, emphasize the adaptive rewards of efforts to be or remain assimilative and perhaps discourage the anxiety-inducing efforts of innovative accommodation. Boys, experiencing a less predictable world where ad hoc strategies are frequently demanded, are pressed more often to reexamine and even to abandon established structures that appear to be ineffective. Over time, success in dealing with such experience would be expected to benefit boys' self-confidence and their freedom to assume the risk of seeking accommodative solutions.

In conceptualizing sex differences in terms of differential orientations toward assimilation and accommodation, it is important to acknowledge that, although clearly the sexes in our culture are socialized in appreciably different ways that may differentially reinforce assimilation and accommodation, it is also the case that the modes of assimilation and accommodation each may be seen as serving an evolutionarily adaptive function, with sexual specialization perhaps a required consequence. Assimilation conserves existing structures, often in creative ways, providing continuity with the past and thus preserving family cultural traditions and historical values. Accommodation results in the creating, sometimes unnecessarily, of new, ahistorical adaptive modes, thus fostering social innovation and cultural change. Both modes of processing experience are adaptively desirable and even necessary. And it is simple to envisage how, given the different requirements of the sexes with respect to enhancing reproductive fitness, females would be oriented more toward the communal mode of assimilation, while males would be oriented more toward the agentic mode of accommodation. Overreliance on either adaptive strategy will prove dysfunctional; complementary invocation of the two modes will be most adaptively effective. Although, from the individual's standpoint, adaptive competence will be greatest when both strategies are readily available, in the larger biosocial context the formation of male-female pairs may result in an adaptive resourcefulness of the reproductive unit often not available to the single individuals involved.

Both biologically ordained and societally shaped, the myriad sex differences observed by psychologists require a conceptual explanation. The heuristic conjecture that boys, more than girls, are ready to invoke the process of accommodation may well assimilate many of the data and other more informal observations on the different ways the sexes tend to resolve disconfirming experiences.

Coda

The present essay has had several aims: to constructively revise a central Piagetian formulation; to specify in more detail than before the steps in a sequence that appears to characterize much of psychological development; to suggest a deeper and more inclusive motiva-

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tional basis for the human tendency to establish, preserve, or reestablish perceptual, knowledge, or action structures sometimes in reality-ignoring ways; and to draw some implications of the several redefinitions and conceptual extensions for the development and functioning of personality.

Within the present perspective, an empirical and a conceptual responsibility will soon have to be faced. To date, cognitive psychology and social psychology have provided extensive and ingenious demonstrations (if not yet a theory) of how already established schemes and heuristics function in people (see Nisbett & Ross [1980] and Taylor & Crocker [1979] for useful reviews). Now, a balancing attention needs to be addressed to an understanding of the processes by which structures develop that offers tenable assimilations of experience. Of particular and dramatic interest for the field of personality will be the study of abrupt, massive, transforming change in perceptual, knowledge, or action structures as the individual accommodatively establishes a sufficient way of encompassing experience. This latter kind of inquiry appears to have gone almost untried in psychology.

Both in past studies of scheme utilization (assimilation) and future studies of scheme reformation (accommodation), the analysis of individual differences can play an important role. As Underwood (1975) has remarked: "individual differences ought to be considered central in theory construction, not peripheral" (p. 129) and may even be the "crucible" for theory testing. Continued ignoring of the presence and implications of individual differences is likely to continue to slow the progression of the science of psychology.

The integral connection of anxiety with the interruption of behavior requires further articulation. At present, the phrase "interruption of behavior" is no more than a convenient, theoretically neutral label for a diverse class of observations and experiences. What is needed now is a more theoretically committed statement regarding the intraindividual and situational parameters involved; the various ways in which the interruption of behavior (anxiety increase) can come about; and the various ways in which the resumption of behavior (anxiety resolution) can be initiated. Such a statement, if systematic, would provide an important advance in our understanding of the dynamics of personality development and functioning.

Reference Notes

1. Block, J. H. Personality development in males and in females: influences of differential socialization. Manuscript submitted for publication, 1980.
2. Carpenter, C. J. Relation of children's sex-typed behavior to classroom and activity structure. Paper presented at the meeting of the Society for Research in Child Development, San Francisco, March 1979.

References

Anderson, S., & Messick, S. Social competency in young children. *Developmental Psychology,* 1974, **10,** 282–293.

Block, J. *The challenge of response sets.* New York: Appleton-Century-Crofts, 1965.

Block, J. H. *Socialization influences on personality development in males and females.* Master lecture cassette tape 15/11. Washington, D.C.: American Psychological Association, 1979.

Block, J. H., & Block, J. The role of ego-control and ego-resiliency in the organization of behavior. In W. A. Collins (Ed.), *Minnesota symposia on child psychology.* Vol. **13.** Hillsdale, N.J.: Erlbaum, 1980.

Crutchfield, R. S. Assessment of persons through a quasi group–situation technique. *Journal of Abnormal and Social Psychology,* 1951, **46,** 577–588.

Fagot, B. I. Influence of teacher behavior in the preschool. *Developmental Psychology,* 1973, **9,** 198–206.

Frenkel-Brunswick, E. Intolerance of ambiguity as an emotional and perceptual personality variable. *Journal of Personality,* 1949, **18,** 108–143.

Harrington, D.; Block, J. H.; & Block, J. Intolerance of ambiguity in preschool children: psychometric considerations, behavioral manifestations, and parental correlates. *Developmental Psychology,* 1978, **14,** 242–256.

Kessen, W. Early cognitive development: hot or cold? In T. Mischel (Ed.), *Cognitive development and epistemology.* New York: Academic Press, 1971.

Lever, J. Sex differences in games children play. *Social Problems,* 1976, **23,** 478–487.

Mandler, G. From association to structure. *Psychological Review,* 1962, **69,** 415–427.

Mandler, G. The interruption of behavior. In D. Levine (Ed.), *Nebraska Symposium on Motivation.* Vol. **12.** Lincoln: University of Nebraska Press, 1964.

Mandler, G. Helplessness: theory and research in anxiety. In C. D. Spielberger (Ed.), *Anxiety:*

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current trends in theory and research. Vol. **2.** New York: Academic Press, 1972.

Mandler, G. *Mind and emotion*. New York: Wiley, 1975.

Mandler, G. The generation of emotion. In R. Plutchik & H. Kellerman (Eds.), *Emotion: theory, research, and experience*. Vol. **1.** *Theories of emotion*. New York: Academic Press, 1980.

Mandler, G., & Watson, D. L. Anxiety and the interruption of behavior. In C. D. Spielberger (Ed.), *Anxiety and behavior*. New York: Academic Press, 1966.

Nisbett, R., & Ross, L. *Human inference: strategies and shortcomings of social judgment*. Englewood Cliffs, N.J.: Prentice-Hall, 1980.

Piaget, J. *Play, dreams and imitation in childhood*. New York: Norton, 1951.

Piaget, J. *The origins of intelligence in children*. New York: International Universities Press, 1952.

Piaget, J. *The construction of reality in the child*. New York: Basic, 1954.

Piaget, J. *Six psychological studies*. New York: Random, 1967.

Piaget, J. Piaget's theory. In P. H. Mussen (Ed.), *Carmichael's manual of child psychology*. Vol. **1** (3d ed.). New York: Wiley, 1970.

Piaget, J. *The development of thought: equilibrium of cognitive structures*. New York: Viking, 1977.

Riegel, K. F. *Psychology, mon amour*. Boston: Houghton Mifflin, 1978.

Rosch, E. Principles of categorization. In E. Rosch & B. B. Lloyd (Eds.), *Cognition and categorization*. Hillsdale, N.J.: Erlbaum, 1978.

Shapiro, D. *Neurotic styles*. New York: Basic, 1965.

Simon, H. A. A behavioral model of rational choice. *Quarterly Journal of Economics*, 1955, **59,** 99–118.

Taylor, S. E., & Crocker, J. Schematic bases of social information processing. In E. T. Higgins, P. Herman, & M. P. Zanna (Eds.), *Toronto Symposium in Personality and Social Psychology*. Vol. **1.** Hillsdale, N.J.: Erlbaum, 1979.

Underwood, B. J. Individual differences as a crucible in theory construction. *American Psychologist*, 1975, **30,** 128–134.

Werner, H. *Comparative psychology of mental development* (Rev. ed.). Chicago: Follett, 1948.

Werner, H. The concept of development from a comparative and organismic point of view. In D. B. Harris (Ed.), *The concept of development: an issue in the study of human behavior*. Minneapolis: University of Minnesota Press, 1957.

Werner, H., & Kaplan, B. *Symbol formation*. New York: Wiley, 1963.

White, R. W. Motivation reconsidered: the concept of competence. *Psychological Review*, 1959, **66,** 297–333.

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